3.3 Unit 3 Investigative and Practical Skills in AS Physics

Candidates should carry out experimental and investigative activities in order to develop their practical skills. Experimental and investigative activities should be set in contexts appropriate to, and reflect the demand of, the AS content. These activities should allow candidates to use their knowledge and understanding of Physics in planning, carrying out, analysing and evaluating their work.

The specifications for Units 1 and 2 provide a range of different practical topics which may be used for experimental and investigative skills. The experience of dealing with such activities will develop the skills required for the assessment of these skills in the Unit. Examples of suitable experiments that could be considered throughout the course will be provided in the Teaching and learning resources web page.

The investigative and practical skills will be internally assessed through two routes;

- Route T – Investigative and Practical skills (Teacher assessed)
- Route X – Investigative and Practical skills (Externally Marked).

Route T – Investigative and Practical skills (Teacher assessed)

The assessment in this route is through two methods;

- Practical Skills Assessment (PSA)
- Investigative Skills Assignment (ISA).

The PSA will be based around a centre assessment throughout the AS course of the candidate's ability to follow and undertake certain standard practical activities.

The ISA will require candidates to undertake practical work, collect and process data and use it to answer questions in a written test (ISA test). See Section 3.8 for PSA and ISA details.

It is expected that candidates will be able to use and be familiar with 'standard' laboratory equipment which is deemed suitable at AS level, throughout their experiences of carrying out their practical activities.

This equipment might include:

Electric meters (analogue or digital), metre rule, set squares, protractors, vernier callipers, micrometer screwgauge (zero errors), an electronic balance, stopclock or stopwatch, thermometer (digital or liquid-in-glass), newtonmeters.

Candidates will not be expected to recall details of experiments they have undertaken in the written units 1 and 2. However, questions in the ISA may be set in experimental contexts based on the units, in which case full details of the context will be given.

Route X – Investigative and Practical skills (Externally Marked)

The assessment in this route is through a one off opportunity of a practical activity.

The first element of this route is that candidates should undertake five short AQA set practical exercises throughout the course, to be timed at the discretion of the centre. Details of the five exercises will be supplied by AQA at the start of the course. The purpose of these set exercises is to ensure that candidates have some competency in using the standard equipment which is deemed suitable at this level. No assessment will be made but centres will have to verify that these exercises will be completed.

The formal assessment will be through a longer practical activity. The activity will require candidates to undertake practical work, collect and process data and use it to answer questions in a written test. The activity will be made up of two tasks, followed by a written test. Only one activity will be provided every year.

Across both routes, it is also expected that in their course of study, candidates will develop their ability to use IT skills in data capture, data processing and when writing reports. When using data capture packages, they should appreciate the limitations of the packages that are used. Candidates should be encouraged to use graphics calculators, spreadsheets or other IT packages for data analysis and again be aware of any limitations of the hardware and software. However, they will not be required to use any such software in their assessments through either route.

The skills developed in course of their practical activities are elaborated further in the How Science Works section of this specification (see section 3.7).

In the course of their experimental work candidates should learn to:

- demonstrate and describe ethical, safe and skilful practical techniques
- process and select appropriate qualitative and quantitative methods
- make, record and communicate reliable and valid observations
- make measurements with appropriate precision and accuracy
- analyse, interpret, explain and evaluate the methodology, results and impact of their own and others experimental and investigative activities in a variety of ways.